

Disproof of TLMC Conjecture 00000001067: the “ $O(1) = 0$ for $p \equiv 3 \pmod{4}$ ” clause for maximal B_2 sets in $\mathbb{F}_p$

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Abstract

TLMC conjecture 00000001067 asserts that the maximal B_2 set of $\mathbb{F}_p$ has size $\lceil \sqrt{p} \rceil + O(1)$ and that the $O(1)$ term equals 0 for $p \equiv 3 \pmod{4}$. Complete exhaustive enumeration shows that for $p = 11 \equiv 3 \pmod{4}$ the maximal B_2 subset of $\mathbb{Z}/11\mathbb{Z}$ has 3 elements, while $\lceil \sqrt{11} \rceil = 4$; likewise $M(19) = 4 < 5 = \lceil \sqrt{19} \rceil$. The exactness clause is therefore false. The counterexample $p = 11$ is fully machine-checked in Lean 4 with zero axioms and zero `sorry`.

1 The conjecture, quoted verbatim

The conjecture text reads, character for character:

“Definition: A B_h set (unique h -fold sums). Conjecture: The maximal B_2 set in $\mathbb{F}_p$ has size $\lceil \sqrt{p} \rceil + O(1)$, and the $O(1)$ term equals 0 for $p \equiv 3 \pmod{4}$ (an exact B_2 result).”

Correspondence. For $h = 2$ in $\mathbb{F}_p$, “unique 2-fold sums” (up to permutation of summands, the only non-degenerate reading) is the Sidon condition: $A \subseteq \mathbb{Z}/p\mathbb{Z}$ is B_2 iff the sums $a + b \pmod{p}$ over all unordered pairs with repetition $a \leq b$, $a, b \in A$, are pairwise distinct. “Maximal” is read as maximum cardinality $M(p) = \max\{|A| : A \subseteq \mathbb{Z}/p\mathbb{Z} \text{ is } B_2\}$, and the $O(1) = 0$ clause asserts $M(p) = \lceil \sqrt{p} \rceil$ exactly for $p \equiv 3 \pmod{4}$. The attack uses precisely these objects; nothing is redefined.

2 The counterexample

A B_2 set of size k in $\mathbb{Z}/p\mathbb{Z}$ has $k(k+1)/2$ distinct unordered pair sums, all lying in a set of p residues. Exhaustive enumeration of *all* subsets gives:

$$M(11) = 3 < 4 = \lceil \sqrt{11} \rceil, \quad M(19) = 4 < 5 = \lceil \sqrt{19} \rceil, \quad 11 \equiv 19 \equiv 3 \pmod{4}.$$

Concretely: for $p = 11$, all $\binom{11}{4} = 330$ four-subsets of $\mathbb{Z}/11\mathbb{Z}$ fail the Sidon condition (each would need its 10 pair sums distinct among 11 residues; none is), while $A = \{0, 1, 3\}$ is B_2 with pair sums $0, 1, 3, 2, 4, 6$ — all distinct mod 11; hence $M(11) = 3$. For $p = 19$, all $\binom{19}{5} = 11628$ five-subsets fail and $\{0, 1, 3, 7\}$ achieves 4 (pair sums $0, 1, 2, 3, 4, 6, 7, 8, 10, 14$, distinct mod 19), so $M(19) = 4$. Since both primes are $\equiv 3 \pmod{4}$, the clause “the $O(1)$ term equals 0 for $p \equiv 3 \pmod{4}$ ” is false; the conjecture’s highlighted “exact B_2 result” does not hold. (The weaker asymptotic part $\lceil \sqrt{p} \rceil + O(1)$ is not contradicted; the deviation is 1 in both examples.) One counterexample suffices: the conjecture is **FALSE**.

Boundary behavior

Exact recomputation for primes $p \equiv 3 \pmod{4}$ shows the exactness fails erratically, and is already forced by the Erdős–Turán bound $M(p) \leq \sqrt{p} + \frac{1}{2}$ at $p = 11, 19$:

p	3	7	11	19	23	31	43	47	59
$M(p)$	2	3	3	4	5	6	6	6	7
$\lceil \sqrt{p} \rceil$	2	3	4	5	5	6	7	7	8
$M(p) = \lceil \sqrt{p} \rceil?$	yes	yes	no	no	yes	yes	no	no	no

3 Lean 4 certificate (zero axioms)

Subsets of $\mathbb{Z}/11\mathbb{Z}$ are encoded as 11-bit masks (bit i set $\iff i \in A$); `isB2 11 n` is the Sidon condition (distinctness of all $a + b \bmod 11$, $a \leq b$). `Main.lean` proves:

- `check11`: an exhaustive check over all $2^{11} = 2048$ masks that no mask with ≥ 4 elements is B_2 , discharged by pure kernel reduction (`rfl`);
- `allRange_spec`: bridge lemma from the Boolean bounded check to a $\forall$ -statement (induction with core lemmas only);
- `no_ge4_B2_Z11`: $\forall n < 2^{11}, \text{isB2 } 11 \ n = \text{true} \rightarrow \text{maskPop } 11 \ n \leq 3$;
- `witness_B2_Z11`: $\{0, 1, 3\}$ (mask 11) is a 3-element B_2 set;
- `disproof_1067`: $11 \% 4 = 3 \wedge (\forall n < 2^{11}, B_2(n) \rightarrow |n| \leq 3) \wedge B_2(\{0, 1, 3\}) \wedge |\{0, 1, 3\}| = 3 \wedge \lceil \sqrt{11} \rceil = 4$.

`Check.lean` runs `#print axioms` on every theorem: each reports “does not depend on any axioms” (no `propext`, `Classical.choice`, `Quot.sound`, `sorryAx`); `native_decide` is not used.

4 Reproduction

1. `python3 reproduce.py` — independent pure-Python recomputation: exhaustive enumeration over all k -subsets of $\mathbb{Z}/p\mathbb{Z}$ yields $M(11) = 3$, $M(19) = 4$, and the boundary table above.
2. `cd lean4 && lake build && lake env lean Check.lean` — compiles the certificate and prints the axiom audit.